When the patrols leave: what Operation Mercurio tells FZS about enforcement against gold mining

Group Giant Anteater · Data for Good · Question 2 · October 2026 · DRAFT

1. THE QUESTION

FZS pays for enforcement and needs to know what it buys. FZS supports SERNANP's operations against alluvial gold mining in the Tambopata–Bahuaja-Sonene landscape (FZS, n.d.). Operations are costly and temporary, so the practical question is: **what happens in the targeted area, and around it, when an operation ends?** Mining could stop (**deterrence**), move next door (**displacement**) or wait and return (**delay**). Inside the target alone these look alike, so we need comparison areas.

Our case: Operation Mercurio, from February 2019 at La Pampa, Peru's largest illegal mining area, in the buffer zone of Tambopata National Reserve (Madre de Dios). **Targeted:** Tambopata buffer zone, 2014–2025. **Comparison:** 14 other Peruvian buffer zones with mining, two rule-based twins abroad, 1-km cells outside the corridor. Amarakaeri is only "less treated": Mercurio II covered all of Madre de Dios in 2019 (MINAM, 2021, p. 33). **Also tracked:** the reserve, the rest of Madre de Dios, and Brazil's Yanomami operation (2023).

2. DATA AND PROVENANCE

Outcome: new mining area per year (ha), MapBiomas Peru Collection 4, class "mining". Buffer zones (course data), check: Tambopata buffer zone 2025 = 20,730 ha, as on the platform; protected areas (our download): Tambopata reserve 2025 = 776.9 ha vs 777; the same maps as rasters cut to our zones: 20,716 vs 20,730 ha. **Context:** World Bank gold price (Aug 2026 = $4,411/oz); MINEM Madre de Dios gold (an artisanal *estimate* until 2019, so not a production series; 2019 = central bank figure); MapBiomas Brazil Col. 11 (indigenous lands 2025 = 39,915 ha); GLAD forest-loss alerts; INGEMMET geology. **Twins:** Amazon Mining Watch (Earth Genome), RAISG, MapBiomas Amazonía Col. 6 (Amazon mining 2023 = 775,746 ha vs 0.78 Mha published). All files in `DOWNLOAD_LOG.csv`; `code/main.py` produces every number and the figure.

3. FINDINGS

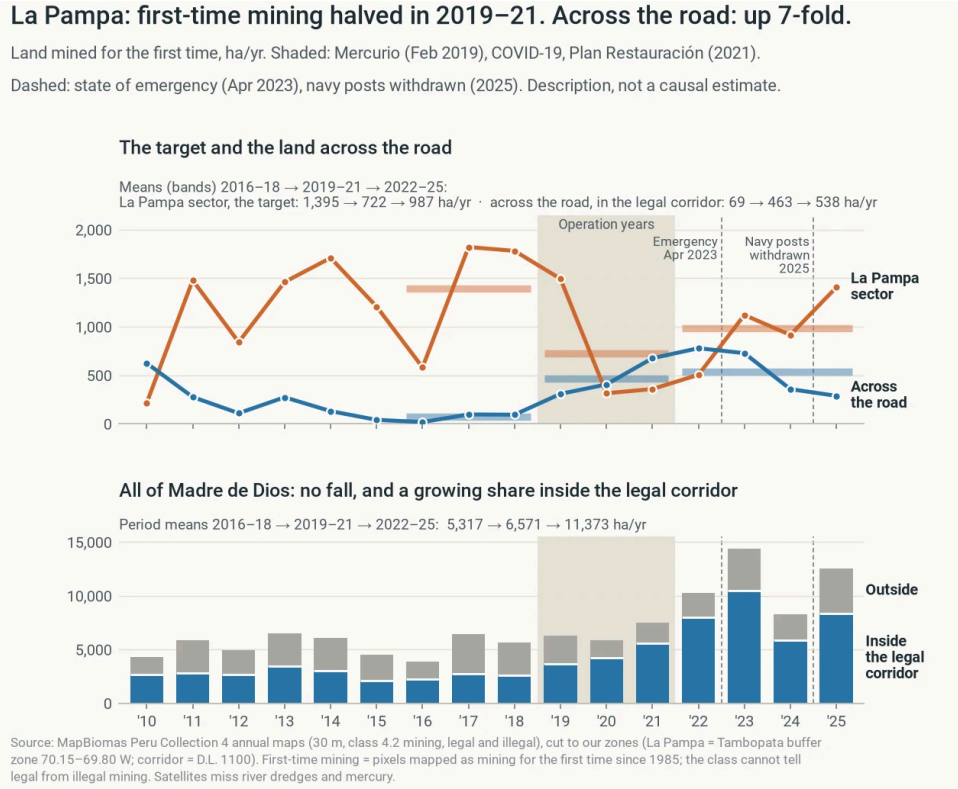


Figure 1. Land mined for the first time per year: La Pampa sector vs the legal corridor across the road (top); all of Madre de Dios, inside vs outside the corridor (bottom). Source: MapBiomas Peru, Collection 4.

During the operation, net new mining in the target collapsed DESCRIPTION. The Tambopata buffer zone fell from 1,640 to 163 ha/yr (2016–18 vs 2019–21) while the less-treated Amarakaeri rose from 289 to 574: a gap of **−1,762 ha/yr**. MAAP saw mining deforestation in La Pampa fall 90% (Finer & Mamani, 2020). Almost half the net

fall is old mining going out of use: *first-time* mining fell less, 2,015 → 1,061 ha/yr (La Pampa sector, in the figure: 1,395 → 722). **Timing:** it stayed near its peak in 2019 and fell only in 2020–21, the COVID years (Dethier et al., 2023, date the halt to 2019).

Large in hectares, ordinary relative to size, smaller with other weights DESCRIPTION. With zone and year effects (absorbing the gold price and COVID as shared shocks), the 2019–21 gap is −1,454 to −1,859 ha/yr and ranks 1 of 15 when each zone in turn gets a fake operation (p = 0.07, this test's floor); scaled to size it ranks 3 of 7. Weighting comparison zones to match Tambopata's past shrinks it: synthetic control −411, synthetic DiD from 2011 −684 ha/yr (both rank 1). Twins abroad and 7 matching estimators all find Tambopata slower (40% of the twins' pace) but none beats chance (placebo p 0.19–0.57), and Tambopata outgrew both twins before 2019. Per 1-km cell La Pampa changes −1.21 ha per cell-year (p = 0.08), but comparison cells hold almost no mining and pre-trends fail (p = 0.005): little more than a before/after.

Therefore CAUSAL, *if the comparison zones show what Tambopata would otherwise have done*, Mercurio and the operations that followed lowered **net** new mining in the target by roughly 411 to 1,859 ha/yr in 2019–21, depending on how the comparison is weighted. This is fragile: it holds in hectares, not relative to size; with one targeted zone the best attainable p is 0.07; and the fall coincides with COVID.

It did not last, and new mining grew across the road DESCRIPTION. In 2022–25 Tambopata averaged 1,829 ha/yr (1,251 without 2023, a spike everywhere), much of it on old pits; the three Madre de Dios buffer zones went 1,941 → 778 → 3,127 ha/yr. Across the highway, in the legal corridor, first-time mining rose from 69 to 463 ha/yr. A spatial regression discontinuity at the road (just north vs just south, before vs after) puts the step +2.1 points of land a year further north (p = 0.08; 0.04–0.98 across bandwidths), counting the fall south and the gain north together; Dethier et al. (2023) also see mining accelerate in permitted concessions north of the highway. The corridor's share of the department's new mining rose from 49% to 72% (MAAP: 74% of 2021–Mar 2024 mining deforestation; Finer & Mamani, 2024) while the total grew (5,317 → 11,373 ha/yr): the corridor's rise exceeds La Pampa's fall, so it is not all miners moving. La Pampa had not run out of gold (21% of its gold-bearing land mined by 2018).

Mining inside the reserve jumped DESCRIPTION: +241 ha in 2025; MAAP reports ~500 ha from late 2025 to Feb 2026, after the navy left the river posts and as gold soared (Pacsi et al., 2026). **Brazil, same method, no evidence** DESCRIPTION: a staggered DiD across 24 territories gives the wrong sign (+0.26 log points, p = 0.44); gold rules changed in 2023 and power is low.

4. WHAT FZS CAN DO WITH THIS

(a) What FZS learns PREDICTION. From this one case, expect a sharp local fall while patrols are present, new mining growing across the nearest legal boundary, a return within about three years (much of it to old pits), and no fall in the region. Asner & Tupayachi (2017) describe a similar regional pattern after the 2012 interdictions (a brief dip, then a near doubling, also inside the reserve), although our maps show no 2012–13 dip in La Pampa. **Fund enforcement as a permanent presence, not a one-off sweep.** Patrol the legal side of the boundary, watch old pits, and staff river posts (the 2025 incursion followed a withdrawal). **Make the corridor a filter:** with a 0% royalty for small producers (Ley 28258, Art. 5) and 9,592 miners registered in formalization but only 145 fully formalized (MINAM, 2021, pp. 55–56), gold-origin tracing there would close an easy outlet.

(b) Why this could mislead, and which way. *Omitted variables:* COVID, the gold price and later waves (2021, 2023) overlap the post-period; year effects remove shared shocks. A COVID-exposure control shrinks the gap to 1,322 of 1,608 ha/yr; adding enforcement money keeps 97%, but money is partly the treatment (a bad control). *Reverse causation:* La Pampa was targeted at its peak, so part of the drop may be a return to normal (overstates); yet every 2019–21 year (≤ 417 ha) fell below every 2011–18 year (≥ 707). *Selection of units:* with one targeted zone standard errors are unreliable (Conley & Taber, 2011), so we rank against fake targets; Amarakaeri received miners (overstates) but had its own operation (understates); the best twin sat in Brazil's weak-enforcement years (overstates). *Measurement:* hectares proxy for miners; the class mixes legal and illegal mining, counts re-mining as new, and **cannot see river dredges or mercury**: MAAP counted 148 → 598 dredges in La Pampa, 2021–23 (Yupanqui et al., 2023). Activity moved to old pits and rivers, which add few hectares. Most biases overstate the fall: **read it as likely an upper bound, and the return as a lower bound**.

(c) What would let us say more. Dated patrol and interdiction records per zone from SERNANP or FZS (enforcement intensity instead of a 0/1 flag); concession and REINFO maps to grade legality pixel by pixel; more operations as cases (Plan Restauración 2021 covered several zones); river-dredge counts and mercury sampling.

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Every team member must re-open what they cite and fill in the access dates. AI is not a source.

APPENDIX A. DATA LOG

See `DOWNLOAD_LOG.csv`: one row per raw file with URL, access date, collection, sha256 and check number.

APPENDIX B. AI OUTPUT FOR PART 4

- **Raw answer:** `docs/brief/ai_part4_raw.md` (prompt as required by the course; tool and date noted there).
- **Our edited version:** part 4(b) above.

What we changed and why:

- The raw answer calls any causal claim "not supported". We kept one causal claim, made it conditional, and backed it with the fake-date and fake-place tests the AI itself suggested.
- We added the mean-reversion check: 2019–21 fell below every year of 2011–18.
- The prompt as sent called Amarakaeri a zone with no targeted operation; we later found Mercurio II covered it (part 1), so our edited version treats it as "less treated" and gives the bias both ways.
- We dropped suggestions we could not carry out: road and geology covariates, and radar.